

[JPJC/2025/Q4]

- 4 JP canteen operates various food stalls in school and wishes to implement a web-based food ordering system to better understand students' food taste and facilitate food ordering. Students can browse dishes online, place food orders, and collect their meals when ready. The system should allow the storage of student details, stall and dish information, and order history.

There are a total of **four** tables, and the table descriptions are as follows:

- `student(studentID, name, phoneNo)`
- `stall(stallID, stallName)`
- `dish(dishID, stallID*, dishName, price, availability)`
- `orderDish(studentID*, dishID*, orderTime, orderDate, quantity)`

The primary key is indicated by underlining one or more attributes.

The attribute indicated with * denoted a foreign key.

It is given that:

- students are allowed to order different dishes in a single transaction. For example, when a student orders **three** different dishes in a single transaction, **three** separate records with the same value for `studentID`, `orderTime`, `orderDate` and `quantity` will be inserted into the table `orderDish`
- the primary keys `stallID` and `dishID` of the tables `stall` and `dish` are auto-incremented integers.
- `orderDate` and `orderTime` in table `orderDish` are represented in the format of `YYYY-MM-DD` and `HH:MM` respectively.

Task 4.1

Create a SQL file named `TASK4_1_<yourclass>_<yourname>.sql` to show the SQL code to create database `canteen.db` with the **four** tables given.

Define the primary and foreign keys for each table.

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Task 4.2

The files `STUDENT.txt`, `STALL.txt`, `DISH.txt` and `ORDERDISH.txt` contain information about the students, stalls, dishes and order history respectively for insertion into the canteen database. Each row in the four files is a comma-separated list of information.

FOR `STUDENT.txt`, information about each student is given in the following order:

NRIC, Name, PhoneNo

Due to the sensitive nature of the National Registration Identity Card (NRIC), and because it can be used to access a range of personal data, it has been decided that a unique identifier for each student will be obtained by combining the last four characters of the NRIC with the first four characters of the student's name, with white spaces removed. All letters in `studentID` will be made uppercase.

For example, a student with `NRIC = "S2345678B"` and `name = "Low Jia Ying"` will be given a `studentID` of `"LOWJ678B"`.

For `STALL.txt`, information about each stall is given in the following order:

`stallID`, `stallName`.

For `DISH.txt`, information about each dish is given in the following order:

`dishID`, `stallID`, `dishName`, `price` and `availability`.

For `ORDERDISH.txt`, information about the dishes ordered by individual students over time.

Write a Python program to insert all information from the **four** files into the **four** separate tables of the canteen database, `canteen.db` based on the table descriptions given above. Run the program.

[8]

Save your program code as

`Task4_2_<yourclass>_<yourname>.py`

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Task 4.3

Write SQL code to show the dish names and prices of all the **available** dishes with a specified stall name. Run this query with the store name "Spicy Delight".

[3]

Save this code as.

Task4_3_<yourclass>_<yourname>.sql

Task 4.4

The stall owners of JP Canteen requested that they would like to know their total sales amount collected on a particular date.

Write a Python program and the necessary files to create a web application. The web application accepts **two** input Stall Name and Date from the web form in separate textboxes. The form design of the web application is shown below:

The form is titled "Calculate Total Sales". It contains two input fields: "Stall Name:" and "Date:". Below these fields is a "Submit" button.

[3]

Save your program code as

Task4_4_<yourclass>_<yourname>.py

With any additional files/subfolders as needed in a folder named

Task4_4_<yourclass>_<yourname>

Run the web application and save the output of the program as

Task4_4_<yourclass>_<yourname>.html

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Task 4.5

Write an SQL query that sums up the total sales amount collected by a stall specified by `stallName` on a given date.

The results of the query should be shown on a webpage in a table that displays the following data:

- Stall ID
- Stall Name
- Date
- Total Sales Collected

The program should return a HTML document that enables the web browser to display a table with required data. The resulting web page is accessible by clicking on the "Submit" button on the web form in **Task 4.4**.

The program should return an HTML document that enables the web browser to display a table with the required data.

[6]

Save your program code as

`Task4_5_<yourclass>_<yourname>.py`

With any additional files/subfolders as needed in a folder named

`Task4_5_<yourclass>_<yourname>`

Run the web application and save the output of the program as

`Task4_5_<yourclass>_<yourname>.html`

Task 4.6

Run your web application by entering the following input into the web form through its "Submit" button.

- Stall Name = "Nasi Padang" [2]
- Date = "2025-07-04"

Run the web application and save the output of the program as

`Task4_6_<yourclass>_<yourname>.html`